

## 6.7 Class Activity: Create a Prototype

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CS XII

Chapter 6: Entrepreneurship in the Digital Age

Lecture 08 of 16

### 6.7 Class Activity: Create a Prototype

#### LEARN IT

**Business idea scenario.** Students often forget **homework deadlines, test dates, and required class items**. A group of students wants to design a simple "**Homework Helper**" or "**ClassBuddy**" reminder system.

The team will **not** immediately build a complete mobile app. They will first **understand the problem, draw a low-fidelity prototype, test it with classmates, and improve it**. This activity is the development cycle of 6.6 in action.

1. Understand → 2. Draw low-fi → 3. Test → 4. Analyse feedback → 5. Improve

#### Step 1: Understand the problem

The team **interviews ten students**. They find that **7 out of 10** students (that is 70%) forget at least one task in a week. Students want **reminders before leaving home**, and some prefer a **very simple design**.

**Table 6.4: Understanding user needs**

| Finding                               | Meaning                     | Design Decision                     |
|---------------------------------------|-----------------------------|-------------------------------------|
| 7 out of 10 forget tasks.             | The problem is common.      | App should show a daily task list.  |
| Reminders needed before leaving home. | Reminder time is important. | Add morning notification option.    |
| Students want simple apps.            | Design must be simple.      | Use large buttons and clear labels. |

#### Step 2: Draw a low-fidelity prototype

The team draws **three screens on paper**: (1) **Today's classes**, (2) **Add homework**, (3) **Show reminder**. **Arrows** show how a user moves from one screen to another.

#### Step 3: Test the prototype

The team gives the paper screens to **five classmates**. The **observer watches silently** to see if users can understand the design by themselves. Users are given one task: "**add one homework task**".

#### Step 4: Analyse feedback

The team looks for **repeated problems** in the feedback to identify what actually needs fixing.

**Table 6.5: Changing feedback into improvement**

| User Feedback                           | Problem Identified             | Improvement                                    |
|-----------------------------------------|--------------------------------|------------------------------------------------|
| "I cannot quickly find Add Homework."   | Button is not visible.         | Make Add Homework button larger and centre it. |
| "I want reminders by subject."          | Reminder is too general.       | Add subject name and due date fields.          |
| "I want to see all tasks for the week." | Only daily view is not enough. | Add a 'Weekly View' button.                    |

## Step 5: Improve the prototype

The team **redraws the home screen** with a **larger Add Homework button**, a **subject list**, and a **weekly view**. This second version is clearer because it is based on real user feedback.

**A prototype has not failed when users give negative feedback; it only fails if the team ignores useful feedback.**

### WORKED EXAMPLE

**Worked example: link the activity to the cycle.**

| Cycle stage | ClassBuddy activity step                                                                 |
|-------------|------------------------------------------------------------------------------------------|
| Design      | Step 1: interview 10 students, make design decisions (Table 6.4).                        |
| Build       | Step 2: draw three paper screens with arrows.                                            |
| Test        | Step 3: five classmates try "add one homework task" while the observer watches silently. |
| Iterate     | Steps 4 and 5: analyse feedback (Table 6.5) and redraw the home screen.                  |

**Why watch silently?** If the observer explains ("tap here"), the team never learns that the button is hard to find. Silence shows the real difficulty.

### REMEMBER IT

**Feedback → Problem → Improvement.** Every comment should be turned into a problem statement and then one clear change, exactly like the three columns of Table 6.5.

### COMMON MISTAKE

**Mistake:** Feeling that negative feedback means the prototype "failed", or changing the design for every single comment.

**Fix:** Negative feedback is useful information. Look for **repeated** problems and fix those first. The prototype only fails if you ignore useful feedback.

### اردو میں سمجھیں

اس سرگرمی میں طلبہ "کلاس بڈی" (ClassBuddy) نامی یاد دہانی نظام بناتے ہیں۔ پہلے دس طلبہ سے بات کر کے مسئلہ سمجھتے ہیں، پھر کاغذ پر تین اسکرینیں بناتے ہیں، پانچ ساتھیوں سے خاموشی سے آزمواتے ہیں، رائے کا تجزیہ کرتے ہیں اور ڈیزائن بہتر کرتے ہیں۔ منفی رائے سے پروٹوٹائپ ناکام نہیں ہوتا؛ ناکامی تب ہے جب مفید رائے کو نظرانداز کیا جائے۔

### CHECK YOURSELF

1. In Step 1, how many students forgot at least one task in a week, and what percentage is that?
2. Name the three paper screens drawn in Step 2.
3. What improvement was made for the feedback "I want to see all tasks for the week"?

Answers: 1. 7 out of 10, which is 70%. 2. Today's classes, Add homework, Show reminder. 3. A 'Weekly View' button was added.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** The class activity exists to:

- a. Make you build and test a prototype, not only define it
- b. Replace the exam
- c. Teach NAND
- d. Teach OSI

**MCQ 2.** Forgotten deadlines is a:

- a. Problem you can prototype
- b. GPU problem
- c. Primary key problem
- d. K-map problem

**MCQ 3.** Five classmates as testers are:

- a. Enough for a class test if you watch silently
- b. A statistically perfect census
- c. Useless
- d. A DoS attack

**MCQ 4.** The observer should:

- a. Watch, not rescue immediately
- b. Click for them
- c. Shout answers
- d. Grade their fashion

**MCQ 5.** You record:

- a. Where they hesitate and what they say
- b. Only a selfie
- c. Only the logo poll
- d. Only Big O

**MCQ 6.** After the activity you still:

- a. Iterate
- b. Ship to Play Store same hour
- c. Skip ethics
- d. Skip the write-up

## Extra short questions

1. Summarise the book's activity scenario.
2. Three paper screens you would draw.
3. How to test without coding.
4. One ethical rule in class tests.

## Extra long question

1. Write a lab report of the activity: idea, 3 screens, 5-user results, 3 changes. Use the cycle words.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Students forget homework/tests/items; team prototypes a reminder.
2. Today's tasks; add task; alarm confirm.
3. Ask a classmate to "use" paper as if it were a phone; you swap sheets.
4. Don't mock testers; don't collect their personal timetable into a public group.

### Long

1. Results: 2/5 missed Add. Changes: bigger button, Urdu, confirm. Iterate. Marks-style headings.